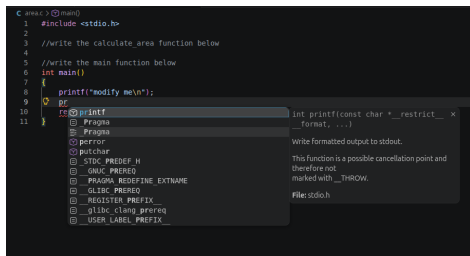


Lab 1: Introduction to C

Matteo Fasiolo

VSCode set up

VSCode comes with auto-suggestion features:



These are useful for experienced programmers, but are detrimental for learners.

The exam is pencil-and-paper, so it's best to disable all these features.

The next slide explains how.

Open VSCode and press Ctrl+Shift+P (Cmd+Shift+P on Macs).

In the text box, type Preferences: Open User Settings.

Copy-paste this code (including brackets) into that file:

```
{
"chat.disableAIFeatures": true,
"editor.inlineSuggest.enabled": false,
"editor.quickSuggestions": false,
"editor.suggestOnTriggerCharacters": false,
"editor.wordBasedSuggestions": "off",
"editor.parameterHints.enabled": false,
"editor.snippetSuggestions": "none",
"editor.tabCompletion": "off",
"editor.acceptSuggestionOnEnter": "off",
"editor.acceptSuggestionOnCommitCharacter": false,
"C_Cpp.autocomplete": "disabled"
}
```

and close that window.

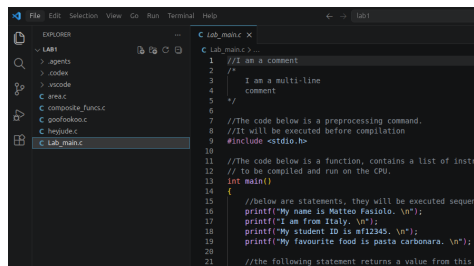
Homework 1

All course material will be on [Github](#).

Download lab1.zip from the Week_by_week/Week_1/lab1 folder and save it to a personal folder.

Extract the files.

Open Visual Studio Code, “open folder”, select the folder which contains the file you have just downloaded.



```
1 //I am a comment
2 /*
3  I am a multi-line
4  comment
5 */
6
7 //The code below is a preprocessing command.
8 //It will be executed before compilation
9 #include <stdio.h>
10
11 //The code below is a function, contains a list of instructions
12 // to be compiled and run on the CPU.
13 int main()
14 {
15     //below are statements, they will be executed sequentially
16     printf("My name is Matteo Fasiolo. \n");
17     printf("I am from Italy. \n");
18     printf("My student ID is mf12345. \n");
19     printf("My favourite food is pasta carbonara. \n");
20
21     //the following statement returns a value from this function
```

Your VS Code window should look similar to this.

Look for `Lab_main.c` and open it.

Compile the `Lab_main.c` file:

- Press `Ctrl + `` to open the command line.
- Use `gcc` to compile the code, as shown in the slides.
- Hint: To hide the command line pressing `Ctrl + `` again.

Execute and observe the program output.

Change the C code in Visual Studio Code, so that it prints out your information, not mine.

Your student ID is the text before your Bristol email address, should be something like “ab23123”.

For example,

My name is [YOUR NAME].

I am from [YOUR HOMETOWN].

My student ID is [STUDENT ID].

My favourite food is [YOUR FAVOURITE FOOD].

Re-compile, execute and observe the output.

Does the program output change?

Write a short comment in your code, listing

- Pros and Cons of high level programming language
- The key components of von Neumann Architecture
- Hint, you can use

```
/*  
to write multi-line  
comments.  
*/
```

Make sure your code compiles and runs!

Homework 2

Open heyjude.c.

Trace the execution of the file step by step using the debugger.

The screenshot shows a Windows 10 desktop with a Visual Studio Code editor window open. The editor displays a C++ program named 'main.cpp' with the following code:

```

1 // Hello World
2 #include <iostream>
3
4 using namespace std;
5
6 int main() {
7     cout << "Hello World" << endl;
8     return 0;
9 }

```

The program is compiled and run, showing the output 'Hello World' in the terminal. The file explorer on the left shows the project structure, including a 'src' folder and a 'main.cpp' file. The status bar at the bottom indicates the file is saved and the editor is in 'C++' mode.

Click on image or [here](#) to play the debugger demonstration.

To enter a function, select **Step Into** (F11) rather than **Step Over** (F10). Step Over executes the function without stepping through its body.

Homework 3

Study the `calculate_circumference` example in the lecture slides, then open `area.c`.

Write a function called `calculate_area` that returns the area of a circle for a given radius.

- Your function should have one input and return one value.

Write a main function that calls `calculate_area` and prints the returned area.

- The output should read: The area of the given circle is X.XXXXXXX.

```
PS C:\Users\songa\Downloads\LabPack\1_functions> & 'c:\Users\songa\.vscode\extensions\ms-vscode.cpptools\bin\WindowsDebugLauncher.exe' '--stdin=Microsoft-MIEngine-In-5rfew023.y5' '--stderr=Microsoft-MIEngine-Error-viaywgkk.fl2' '--pid=Microsoft-MIEngine-Pid-gt' 'LabPack\util\msys64\mingw64\bin\gdb.exe' '--interpreter=mi'
The area of the given circle is 7.0685834.
```

Homework 4

- Open `goofookoo.c`.
- **Do not run the code yet.**
- Read the code and write down the program's output on paper.
- Now run the code and check your answer. Did you get it right? Use the debugger to trace the execution.
- This was a question in 2024 written exam.

Submission

Compress your solution to homework 1 and 3 in a Zip file.

The zip file should be called YOUR_STUDENT_ID.zip.

Your student ID should be something like “ab23123”.

Upload the zip file [on Blackboard](#).

Before submitting, make sure your code compiles and runs!